

```

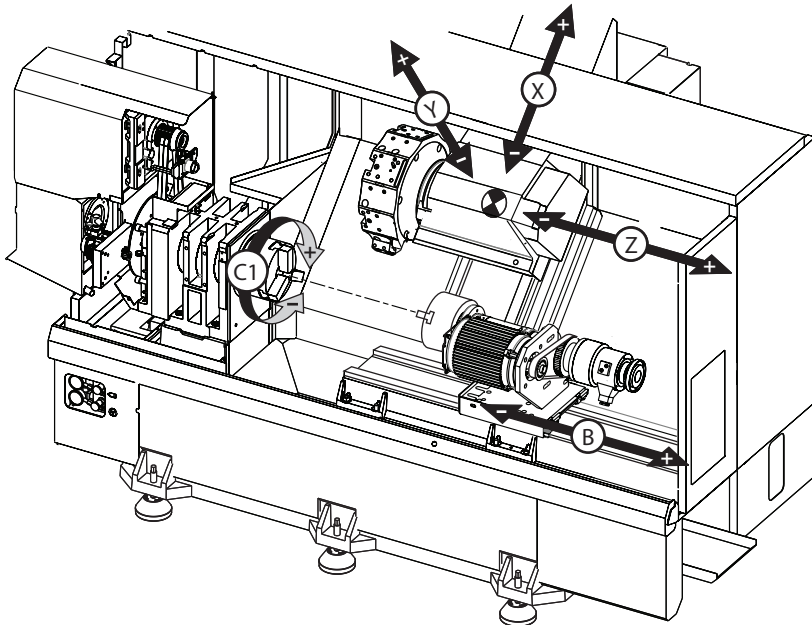
M08 (Coolant on) ;
G00 Z-0.75 (Rapid to Z depth) ;
(BEGIN CUTTING BLOCKS) ;
G75 X1.5 I0.25 F6. (Begin G75 on 1st hole) ;
G00 C180. (Rotate C axis to new position) ;
G75 X1.5 I0.25 F6. (Begin G75 on 2nd hole) ;
G00 C270. (Rotate C axis to new position) ;
G75 X1.5 I0.25 F6. (Begin G75 on 3rd hole) ;
(BEGIN COMPLETION BLOCKS) ;
G00 Z0.25 M09 (Rapid retract, coolant off) ;
M155 (Disengage C axis) ;
M135 (Live tool off) ;
G18 (Return to XZ plane) ;
G53 X0 (X home) ;
G53 Z0 (Z home) ;
M30 (End program) ;

```

6.4 Dual-Spindle Lathes (DS-Series)

The DS-30 is a lathe with two spindles. The main spindle is in a stationary housing. The other spindle, the “secondary spindle”, has a housing that moves along a linear axis, designated “B”, and replaces the typical tailstock. You use a special set of M-codes to command the secondary spindle.

F6.3: Dual Spindle Lathe with an Optional Y Axis



6.4.1 Synchronized Spindle Control

Dual-spindle lathes can synchronize the main and secondary spindle. This means that when the main spindle receives a command to turn, the secondary spindle turns at the same speed, in the same direction. This is called Synchronous Spindle Control (SSC) mode. In SSC mode, both spindles accelerate, maintain speed, and decelerate together. You can then use both spindles to support a workpiece at both ends for maximum support and minimum vibration. You can also transfer the workpiece between the main and secondary spindle, effectively doing a “part flip” while the spindles continue to turn.

There are two G-codes associated with SSC:

G199 activates SSC.

G198 cancels SSC.

When you command G199, both spindles orient before they accelerate to the programmed speed.



NOTE:

When you program synchronized dual spindles, you should first bring both spindles up to speed with M03 (for the main spindle) and M144 (for the secondary spindle) before you command G199. If you command G199 before commanding spindle speed, the two spindles attempt to stay synchronized while they accelerate, causing acceleration to take much longer than normal.

If SSC mode is in effect, and you press **[RESET]** or **[EMERGENCY STOP]**, SSC mode remains in effect until the spindles stop.

The Synchronized Spindle Control Display

The spindle synchronization control display is available in the **CURRENT COMMANDS** display.

The **SPINDLE** column gives the main spindle status. The **SECONDARY SPINDLE** column gives the secondary spindle status. The third column shows miscellaneous status. On the left is a column of row titles:

G15/G14 - If G15 appears in the **SECONDARY SPINDLE** column, the main spindle is the leading spindle. If G14 appears in the **SECONDARY SPINDLE** column, the secondary spindle is the leading spindle.

SYNC (G199) - When G199 appears in the row, spindle synchronization is active.

POSITION (DEG) - This row shows the current position, in degrees, of both the spindle and the secondary spindle. Values range from -180.0 degrees to 180.0 degrees. This is relative to the default orientation position of each spindle.

The third column indicates the current difference, in degrees, between the two spindles. When both spindles are at their respective zero marks, then this value is zero.

If the third column value is negative, it represents how much the secondary spindle currently lags the main spindle, in degrees.

If the third column value is positive, it represents how much the secondary spindle currently leads the main spindle, in degrees.

VELOCITY (RPM) - This row shows the actual RPM of the main spindle and the secondary spindle.

G199 R PHASE OFS. - This is the programmed R value for G199. This row is blank when G199 is not commanded; otherwise it contains the R value in the most recently executed G199 block.

Refer to page **326** for more information on G199.

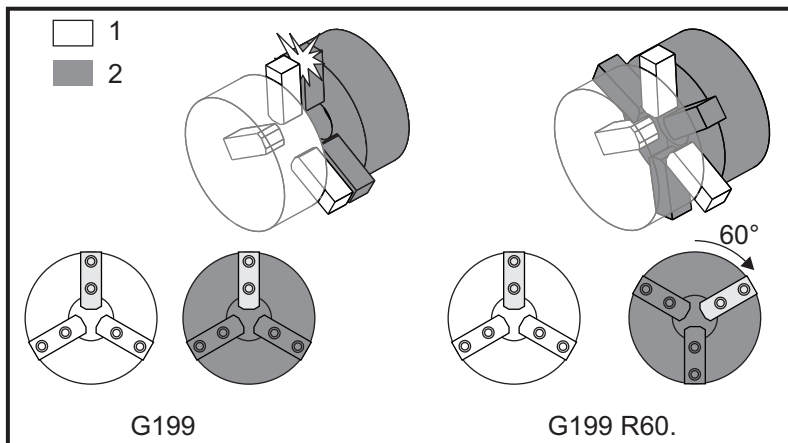
CHUCK - This column shows the clamped or unclamped status of the work holding (chuck or collet). This row is empty when clamped, or shows “UNCLAMPED” in red when the work holding is open.

LOAD % - This shows the current load percent for each spindle.

R Phase Offset Explained

When dual lathe spindles are synchronized, they orient, then rotate at the same speed with their home positions stationary relative to each other. In other words, the relative orientation you see when both spindles are stopped at their home positions is preserved as synchronized spindles rotate.

You can use an R value with **G199**, **M19**, or **M119** to alter this relative orientation. The R value specifies an offset, in degrees, from the following spindle’s home position. You can use this value to allow the chuck jaws to mesh during a workpiece hand-off operation. Refer to Figure **F6.4** for an example.



Finding a G199 R Value

To find an appropriate G199 R value:

1. In **MDI** mode, command an **M19** to orient the main spindle and an **M119** to orient the secondary spindle.
This establishes the default orientation between the spindles' home positions.
2. Add an **R** value in degrees to the **M119** to offset the secondary spindle's position.
3. Check the interaction between the chuck jaws. Change the **M119 R** value to adjust the secondary spindle position until the chuck jaws interact correctly.
4. Record the correct **R** value and use it in the **G199** blocks in your program.

6.4.2 Secondary Spindle Programming

The program structure for the secondary spindle is the same as that for the main spindle. Use **G14** to apply main spindle M-codes and canned cycles to the secondary spindle. Cancel **G14** with **G15**. Refer to page **275** for more information on these G-codes.

Secondary Spindle Commands

Three M-Codes are used to start and stop the secondary spindle:

- **M143** starts the spindle forward.
- **M144** starts the spindle in reverse.
- **M145** stops the spindle.

The **P** address code specifies the spindle speed, from 1 RPM to maximum speed.